

Diffusion-MoE Phase 2 Training Investigation

Findings report — all-layers training run, bugs found, and current state

Prepared for review · *diffusion-moe project*

Executive Summary

Following Phase 1's diagnostics (`phase1_findings_report.md`) and the bounded single-layer pilot fine-tune, this phase asked the question those tools structurally couldn't answer: **what actually happens if every layer of the project's own from-scratch architecture is a `DiffusionMoELayer` , trained end-to-end, not just diagnosed or spliced into a frozen pretrained backbone?**

The honest answer is: six real, previously-undiscovered bugs, one badly-miscalibrated hyperparameter, and — after fixing all of them — clean, complete training runs that demonstrate the architecture can learn, route without permanently collapsing, and finish without crashing. That is real progress, not a finished result: a same-budget dense-model control, repeated across two seeds, decisively and consistently outperformed the all-layers diffusion-MoE configuration on quality, cost, *and* reliability (§6.4–6.5). See §7 for the full picture and what would change that conclusion.

1. **NaN divergence at step ~199** — `NystromDiffusionMap` 's kernel bandwidth (`eps_`) is fit once and frozen between landmark refits; once training moves the embedding distribution far enough, a token can drift outside every landmark's kernel support, and the unguarded division silently produces NaN, which `clip_grad_norm_` cannot repair. Fixed with both a defensive guard and — the deeper fix — a live `eps_` refresh between refits (§2.1).
2. **Checkpoint resume crashes a real MoE layer** — `NystromDiffusionMap` 's fitted state lives on a plain Python object, not an `nn.Module` , so it silently does not survive `state_dict()` save/load. Any resume off the exact refit-schedule boundary crashed outright. Found running the project's own `expert_attribution.py` tool against a real checkpoint, not by inspection (§2.2).
3. **No shared/always-active expert** — every FFN computation was contingent on sparse top-k routing; nothing had guaranteed, routing-independent gradient signal. Added a mandatory `shared_expert` , architecturally identical to a routed `ExpertFFN` , applied unconditionally outside the router (§3).
4. **Disk-full crash near the finish line** — unbounded checkpoint retention filled a 60GB instance's disk with ~5.7GB files, corrupting the final checkpoint write. Fixed with automatic pruning (§4).
5. **A catastrophic initialization bug** — `task_loss` started at ~823 nats against a random-guessing baseline of $\ln(32000)=10.37$, and never recovered below ~60 nats across any run, no matter how long training continued. Root-caused to two compounding, previously-unaddressed initialization gaps (no custom init existed anywhere in the model) and fixed; confirmed both in isolation and in a real training run (§5). This is the most consequential finding in this phase — everything observed in earlier runs (§6) was training on top of this bug.
6. **`mu_load` (the load-balance loss weight) was miscalibrated by ~2 orders of magnitude** relative to `task_loss` 's gradient once the loss scale was actually correct — measured directly, not guessed, and retuned (§6.2).
7. **`cfg.seed` was a no-op** — never applied to torch's global RNG anywhere in this pipeline (only `data.seed` , controlling shuffle order, was wired through), so model weight initialization — the dominant source of run-to-run variation — drew from whatever unseeded state the process happened to start in. Fixed, and what made the seed-repeat check in §6.5 meaningful in the first place (§9).

The final MoE run (`graceful-universe-21`) completed its full 2,441-step / 20M-token budget cleanly: zero NaN, `task_loss` ending at 7.08 (below the random baseline), all 24 layers' `load_loss` ending near zero (down from 7 of 24 near-total collapse mid-run), and `val_pp1` finally moving off its `exp(20)` safety cap for the first time in this investigation (7,286 → 5,081 → 35,294 spike → 16,158 → 3,245 → 2,239 → 974). That is real progress relative to every earlier run in this phase — but a same-budget **dense-model control run** (`peachy-cloud-22` , §6.4), run immediately afterward, decisively beat it: `task_loss=4.80` , `val_pp1=93.2` , in ~15x less wall-clock time. A second seed of both configs (§6.5) confirmed this wasn't one lucky/unlucky comparison — dense stayed essentially unchanged (sub-1% variance), while the MoE run got worse on the second seed (`val_pp1=1,478` , wall-clock 425 min) rather than better. See §7.

§8 (added after the above was first written) goes further and identifies why. A single, data-driven selective layer (§8.1) closes most of the all-layers gap on its own — but investigating *why* the routed layer still lagged dense surfaced a real bug (`centroid_refresh_steps` was silently measured in micro-batches, refitting twice as often as configured, §8.2) and, more importantly, a structural finding: `task_loss` never recovers to its pre-refit level within a full refit-to-refit window (§8.3) — the diffusion map is fighting a representation that's continuously reshaped by the same gradient descent the routing depends on, not just suffering a brief coordinate-sign hiccup. A targeted fix for the latter (§8.4) helped only at the margin. The decisive test: `SwitchMoELayer` — already in this codebase, a standard continuously-trained linear router with no periodic refit at all — lands within a few percent of dense at the identical layer selection (§8.6), and two follow-up runs (§8.8) confirm this generalizes: Switch gets *closer* to dense when scaled to all layers (the opposite of diffusion, which got 4.6x worse), and is seed-robust like dense (sub-1% seed variance) rather than seed-sensitive like diffusion (44–52%). **§8.9 then found that comparison itself wasn't fair:** dense's baseline config ran 2.7x more active FFN compute per token than Switch the entire time. Compute-matched, dense and Switch are statistically *tied* (`val_pp1` 100.33 vs 100.4) — Switch isn't beating diffusion by being a better architecture at equal resources, it's tied with dense while using 1/2.7 the compute and diffusion is losing to both. **This phase's honest bottom line, updated:** the infrastructure now works; the diffusion-geometric router's underperformance is specifically the periodic-hard-refit design (not sparse routing or this project's scale, per §8.6/§8.8); and separately, sparse MoE (Switch) hasn't yet demonstrated a quality advantage over a compute-matched dense model at this scale/budget — only compute parity, at 3x the parameter cost (§8.9). Per-project decision at the close of this phase: further diffusion-router development is on hold pending a real structural fix to the moving-target problem (§8.3); whether Switch-style MoE earns its complexity is a separate, still-open question pending a larger scale point.

1. Background and Motivation

Phase 1's diagnostics and the bounded pilot fine-tune (`phase1_findings_report.md` §3.4–3.6) each tested one layer at a time, spliced into an otherwise-frozen pretrained Mistral-7B backbone. That structurally cannot answer a sharper question raised during review: *if a layer's router keeps forcing collapse onto one expert during real training, is that proof the layer can't benefit from diffusion routing at all, or just proof that one frozen-backbone, one-layer splice hasn't given the router anything to differentiate?* The only way to find out is to train the project's own from-scratch architecture with every layer routed, and watch. The user's explicit direction: *"I would let all layers use diffusion MoE and see what happens."*

This phase covers that run, the infrastructure it required (per-layer loss breakdown, checkpoint resume, expert-attribution analytics — see `methodology.md` §5 and §6 for the durable reference material), the bugs it surfaced, and the final result.

All runs in this section use the project's own 300M/24-layer from-scratch architecture (`configs/model/300m.yaml`), trained on `wikitext-103-raw-v1` via local parquet files (bypassing the

recurring flaky-CDN Hub-streaming stall — see `methodology.md §4`), on a single rented RTX 3090 instance (vast.ai).

2. Bug: NaN Divergence, and the Deeper Fix Behind It

2.1 Root cause: a frozen kernel bandwidth under a moving distribution

The first all-layers run diverged to NaN at step 199, confirmed via an explicit `RuntimeWarning: invalid value encountered in divide` in `geometry/nystrom.py`. `NystromDiffusionMap.transform()`'s Nyström extension formula divides by each new point's total kernel mass to the (frozen) landmarks:

```
K_alpha = K_new / (d_alpha_new[:, None] * self._d_alpha_landmarks[None, :])
```

`eps_` (the kernel bandwidth) is derived once from the landmarks' own spread at the last refit (`fit()`), only run every `centroid_refresh_steps`) and then frozen until the next one. If a token's embedding drifts far enough from that frozen reference — plausible once `centroid_separation_loss` has been actively pushing centroids apart for hundreds of steps — its kernel value to every landmark can underflow to exactly `0.0` in float64 (`exp(v)` underflows to the literal bit pattern `0.0` below roughly `v < -745`, not a small-but-nonzero value). The resulting `0/0` division produced `NaN`, which propagated through `task_loss / load_loss`, and — critically — `torch.nn.utils.clip_grad_norm_` cannot repair a NaN gradient (the norm of a NaN tensor is itself NaN), so every parameter update after that point silently corrupted the entire model. Training continued to "run" for ~40 more steps on fully poisoned weights before crashing outright on an unrelated downstream `KMeans` call (sklearn's input validator rejects NaN explicitly; the raw numpy division earlier did not).

Immediate fix: guard the division (`d_alpha_new_safe`, mirroring the existing `row_sums` guard a few lines below) so a token with zero kernel mass gets diffusion coordinates at the origin (maximal uncertainty) instead of NaN.

2.2 The real fix: live-refreshing the kernel bandwidth between refits

The guard alone treats a symptom. The actual staleness problem — `eps_` describing a step-0 snapshot of the embedding distribution, growing more wrong the longer training runs between refits — remains. Fixed by having `transform()` re-derive `eps_` from the *current* batch's actual distance to the (still-frozen) landmarks on every call, reusing the distance matrix already computed for the Nyström kernel (cheap — no new k-means).

This surfaced two further correctness requirements, both needed for the fix to be mathematically sound rather than merely avoid a crash:

- **eigenvectors / psi_landmarks_ must be re-derived alongside `eps_`** (extracted into a shared `_fit_landmark_spectrum`, called by both `fit()` and the refresh path): Nyström's defining correctness property — that applying the extension formula to the landmarks themselves must exactly reproduce their direct diffusion coordinates — only holds when the landmark spectrum and the new-point extension share one consistent kernel. Caught by `test_nystrom_quality_exact_reproduction_on_landmarks` when a first attempt (refreshing only `eps_` in isolation) broke it (0.0042 error against a $1e-4$ tolerance).
- **diffusion_eigenvectors's ARPACK solve needed a seeded starting vector.** It previously ran once per landmark set, so its default unseeded random start never mattered. Once it could rerun every step, an unseeded start would make the diffusion-coordinate basis jitter step-to-step from

solver randomness alone — independent of any real drift in the data, i.e. the opposite of what this fix was for.

- **fit_transform** 's own internal **transform()** call must skip the refresh (new `_refresh_eps` parameter): it runs immediately after `fit()` on the same data, so there is no staleness yet to correct, and refreshing anyway would silently override `fit()` 's landmark-based heuristic and redo the eigensolve twice on every refit step for no benefit.

Cost: this is not free, but it is cheap relative to what it replaces. `eps_` 's refresh reuses an already-computed distance matrix; the re-derived eigensolve is over just the `n_landmarks × n_landmarks` matrix (32–128 points), not another k-means clustering over the full token batch. Measured directly in production: steady-state per-step time rose from ~5.55s to ~7.0–7.3s (roughly 25–30%), on top of an already CPU-bound geometry pipeline (this entire subsystem runs on `z.detach()` 'd numpy/sklearn, never touching the GPU).

3. Feature: A Mandatory Shared Expert, Outside the Router

Once every layer was routed and training could be observed cleanly, a striking pattern emerged (visible even before the initialization bug was found, §5): `load_loss` stayed elevated across nearly all 24 layers simultaneously — not one collapsed outlier, a near-universal, if partial, imbalance (mean `load_loss` 3.21 across all layers at one snapshot, only 1 of 24 layers below 1.0). This raised the mechanistic question motivating this whole run: with every FFN computation contingent on sparse top-k routing, nothing in the network had guaranteed, routing-independent gradient signal — common/generic computation had to be either redundantly relearned by whichever expert won that step's routing lottery, or not learned reliably at all if routing was noisy.

This is a known pattern in production MoE architectures (DeepSeekMoE's "shared expert isolation," similarly Qwen2-MoE), added here as `DiffusionMoELayer.shared_expert` : architecturally identical to a routed `ExpertFFN` (same width, same `overlap_factor`), applied unconditionally to every token, every step — no gate value, no centroid, entirely outside the diffusion router's influence. This adds real active compute per token (`top_k + 1` experts fire instead of `top_k`), not a reallocation of existing capacity.

Verified with a direct behavioral test

(`test_shared_expert_contributes_even_when_routing_contributes_nothing`): with the routed dispatch path monkeypatched to contribute exactly zero, the layer's output still differs from the plain attention residual — the shared expert's contribution cannot be made routing-contingent by construction.

4. Bug: Disk-Full Crash Near the Finish Line

A run that had otherwise reached step 2,400 of 2,441 (98.3%) crashed on its own checkpoint save: the instance's 60GB disk filled completely (8 accumulated checkpoints × ~5.7GB each), and the final write was cut off mid-file (`PytorchStreamWriter failed writing file` ; the resulting `step_2400.pt` was truncated to ~2.97GB against an expected ~5.68GB — corrupt, unusable). The last *valid* checkpoint (`step_2100.pt` , saved before the disk filled) was confirmed to load cleanly.

Fixed with automatic checkpoint retention in `Trainer` : after each numbered checkpoint save, all but the most recent `keep_last_n_checkpoints` (default 2, configurable via `training.keep_last_n_checkpoints`) are deleted. `best.pt` is a separate file and always kept regardless. `<= 0` disables pruning (old, unbounded behavior) for anyone who wants it.

5. The Initialization Bug: Task_Loss Was Never Actually Calibrated

5.1 The anomaly

Every run before this fix showed the same signature: `task_loss` starting around 823 nats (step 1) and never recovering below roughly 60 nats, however long training continued. `val_ppl ( exp(min(avg_nll, 20.0)) )` was pinned at its safety cap — 485,165,195.4 exactly — at *every single logged evaluation, across every run, all the way to the end of a 2,400-step run*. This is a real red flag independent of routing behavior: this project's tokenizer (Mistral's, 32,000 tokens) gives a random-guessing baseline of $\ln(32000) \approx 10.37$ nats — the cross-entropy a properly-initialized, completely untrained model should sit near. A trained model that never gets within even a factor of 2 of that baseline, in either direction, is not "undertrained" — something is numerically broken.

5.2 Root cause, confirmed by direct measurement

No custom weight initialization existed anywhere in the model — every linear layer and the token embedding relied entirely on PyTorch's defaults. Two compounding gaps, confirmed empirically (not just inferred) by hooking every block of a freshly-constructed 24-layer model and measuring residual-stream norms and final logit magnitude directly:

1. **token_embedding used nn.Embedding's default init — Normal(0, 1), std=1.0** — far too large for a transformer embedding table, and especially costly because `tie_embeddings=True` by default makes this same oversized matrix double as the LM-head unembedding projection, directly inflating logit magnitude. This was the *dominant* cause: fixing only the second gap below (residual-output scaling) left final logits unchanged — actually slightly worse (`max abs` 828 → 1071) — until this was also fixed.
2. **No depth-aware scaling on residual-branch output projections** (attention's `out_proj`, every FFN's `down_proj` — dense TransformerBlock, routed ExpertFFN, and the new shared_expert alike). Every serious transformer implementation (GPT-2, LLaMA, nanoGPT) scales these down by $1/\sqrt{2 \times n_layers}$ at init specifically because residual-stream variance otherwise compounds with depth. Confirmed directly: residual-stream norm grew 32 → 44 over 24 layers unfixed; a controlled, smooth 1.0 → 7.1 fixed.

Combined fix: `token_embedding` initialized to the standard `std=0.02` (GPT-2/nanoGPT convention); every `out_proj` / `down_proj` weight scaled by $1/\sqrt{2 \times n_layers}$ after construction (matched on the Linear's own attribute name, so it applies uniformly regardless of which module — dense, routed expert, or shared expert — it's nested in).

Verified twice. In isolation: a fresh random-init model's cross-entropy against random labels dropped from ~1,014 to **10.51** (essentially exactly $\ln(32000)=10.37$). In real training (`vital-microwave-20` , the first run with this fix): step 1 `task_loss` = 10.61 , and a clean, sane trajectory afterward (10.61 → 9.03 → 8.47 by step 26) — nothing resembling any earlier run's trajectory.

5.3 What this means for everything observed before this fix

`resilient-brook-18` and `hardy-lake-19` (§6) both trained on top of this bug. Their qualitative conclusions (per-layer collapse patterns, shared-expert comparison) are likely still valid as *relative* signals within those runs, but their absolute loss numbers should not be treated as meaningful, and the shared-expert-vs-not comparison in §6.2 should be re-run now that the loss scale is correct before drawing any firm conclusion from it.

6. Run-by-Run Summary

Run (wandb)	Init fix	Shared expert	mu_load	Outcome
resilient-brook-18	No	No	0.01	task_loss plateaued 60–110 nats; val_ppl permanently capped; killed to add the shared expert
hardy-lake-19	No	Yes	0.01	Modest improvement over the above at matched steps (task_loss=90.3 vs 100.3 @ step 600) but still broken-scale; crashed on disk-full at step 2,400/2,441 (§4)
vital-microwave-20	Yes	Yes	0.01	Step 1 task_loss=10.61 (correct!), but 18 of 24 layers collapsed to load_loss≈7.0 (the exact CV ² ceiling) by step ~600; task_loss climbed back to 11.5 — worse than its own starting point; killed for mu_load retuning
graceful-universe-21	Yes	Yes	2.0	Completed cleanly — see §6.3
peachy-cloud-22 (dense control, layers_to_replace=[])	Yes	n/a	n/a	Same budget, same init. Beat graceful-universe-21 decisively — see §6.4
sunny-surf-23 (dense, seed=123)	Yes	n/a	n/a	Near-identical to peachy-cloud-22 — see §6.5
cool-pyramid-24 (MoE, seed=123)	Yes	Yes	2.0	Completed cleanly, but meaningfully worse and slower than graceful-universe-21 — see §6.5

6.1 The collapse dynamic got worse, not better, once the loss was fixed

vital-microwave-20 is the sharpest result in this phase. Fixing the initialization did not just fix the loss scale — it let gradients flow cleanly for the first time. task_loss 's gradient became a strong, well-calibrated signal instead of one dominated by first fixing a catastrophic starting point, which made the router-collapse shortcut (route everything to one expert per layer, reducing the effective optimization problem) more attractive to find, not less. 18 of 24 layers reached essentially the exact load_loss ceiling ($n_experts - 1 = 7.0$) within ~600 steps. This is a direct escalation of the question this whole run was designed to answer: collapse is not a rare pathology in one unlucky layer, it is close to the default outcome once the network can actually learn efficiently, unless something resists it hard enough.

6.2 Measuring, not guessing, the fix

Rather than guess a new mu_load , the relative gradient pull of task_loss vs. load_loss on the shared ExpertCentroids parameter was measured directly on a fresh, correctly-initialized model (the parameter both losses actually compete to move, since load_loss 's gradient is confined to centroids — Psi_t is detached before the routing-loss computation, so it never reaches attention/backbone weights directly). Raw ratio: task_loss's pull was only **1.86×** load_loss's — a fair fight. The old mu_load=0.01 shrank that to an effective **186×**, leaving load_loss almost no real ability to resist collapse — a value very likely tuned against the old, catastrophically-mis-scaled task_loss and never revisited once §5's fix corrected it. Set to mu_load=2.0 , targeting an effective ratio near 1.

6.3 The final run

`graceful-universe-21` completed the full 2,441-step budget cleanly — zero NaN, `state=finished` . At the halfway point (step ~296), the picture was a real but partial improvement: 7 of 24 layers near collapse, versus 18 of 24 in the unfixed- `mu` run at a comparable point — down ~60%, not eliminated. By the *end* of training, the picture resolved further: all 24 layers' `load_loss` near zero (best 0.008, most below 0.0005) — though the path there was genuinely volatile, not a clean monotonic decrease (`load_loss` oscillated between 0.45 and 3.6 across steps 1,200–2,200 before settling as the learning rate decayed toward its floor; see §7 for why this specific ending should not yet be over-interpreted).

`task_loss` ended at **7.08** — below the random baseline, genuine learning. `val_ppl` moved for the first time in this entire investigation, off the `exp(20)` cap:

```
step 300: 7,286
step 600: 5,081
step 900: 35,294  <- real spike, not a fluke
step 1200: 16,158
step 1800: 3,245
step 2100: 2,239
step 2400: 975   <- best of the run
```

Non-monotonic and still high in absolute terms for a 300M-parameter model (even on a small budget), but a real, measured signal that the model is generalizing at all — something no earlier run in this entire investigation had shown.

6.4 The dense-model control, and a decisive negative result

Immediately after `graceful-universe-21` completed, the same config was rerun with `layers_to_replace=[]` (dense baseline, identical data/budget/token count/hyperparameters otherwise) — the missing control flagged in every version of §7 until now. Same starting point confirmed (`task_loss=10.59` at step 1 vs. the MoE run's `10.61` , so the initialization fix applies uniformly and this is a fair comparison):

	Dense (<code>peachy-cloud-22</code>)	Diffusion-MoE, all layers (<code>graceful-universe-21</code>)
Final <code>task_loss</code> (step 2441)	4.80	7.08
Final <code>val_ppl</code>	93.2	974.6
<code>val_ppl</code> trajectory	Smooth, monotonic: 274 → 200 → 156 → 131 → 100 → 95 → 93	Noisy, huge spike at step 900: 7,286 → 5,081 → 35,294 → 16,158 → 3,245 → 2,239 → 975
Wall-clock, same 2,441 steps	20.1 min	294.5 min (~4.9 hours, ~15x longer)

The dense model wins decisively on every axis, at matched token budget. The time gap is almost entirely the CPU-bound Nyström/eigensolve pipeline (~7 s/step MoE vs. ~0.49 s/step dense — matches the ~15x wall-clock ratio directly), not GPU compute. This is not even a "worse quality, but cheaper active compute" tradeoff in MoE's favor: the diffusion-MoE model's active FFN width per token (`top_k+1=3` experts × 512

each = 1,536) is *smaller* than the dense model's full 4,096-wide FFN — less active capacity, ~15x the wall-clock cost, and a meaningfully worse result.

This is a real, decisive answer, and it is unfavorable to the current approach. Nothing in this investigation shows the added complexity of diffusion-routed MoE buying anything at this scale/budget. The standard framing for why sparse MoE is worth its complexity (Clark et al. 2022's scaling-law work, referenced earlier in this investigation) is specifically about quality-per-active-FLOP *at scale* — 20M tokens on a 300M-parameter model is nowhere near the regime that literature has actually demonstrated an advantage in. This result is consistent with that: it does not show diffusion-MoE is a bad idea in general, but it does mean this specific comparison offers no evidence *for* it, and meaningful evidence against training it further at this scale without a specific, articulated reason to expect the gap to close only at larger scale.

6.5 Seed-repeat check: dense is consistent, diffusion-MoE is not

Both §6.4 configs were rerun with `seed=123` (a new fix this phase: `cfg.seed` was previously never actually applied to torch's global RNG — only `data.seed`, controlling shuffle order, was wired through, meaning every prior run's model initialization was drawn from whatever unseeded state the process happened to start in, not a controlled value at all — see §9).

	seed=42	seed=123	Spread
Dense task_loss	4.80	4.80	~0%
Dense val_pp1	93.2	92.8	<1%
Dense wall-clock	20.1 min	20.0 min	~0%
MoE task_loss	7.08	7.44	~5%
MoE val_pp1	974.6	1,477.5	~52%
MoE wall-clock	294.5 min	424.6 min	~44%
MoE load_loss (final)	0.00044	0.00020	both ≈0

Two things confirmed here. First, §6.4's dense-beats-MoE result is not a fluke of one lucky/unlucky seed pair — dense is close to seed-invariant, and the *worse* of the two MoE seeds (seed=123, val_pp1=1,478) is still over 15x worse than *either* dense run. Second, and a genuinely new finding rather than just confirmation: diffusion-MoE is not just worse on average, it is meaningfully *less consistent* than dense, on both quality and cost — a 52% perplexity swing and a 44% wall-clock-time swing between two runs that differ only in random seed, versus dense's sub-1% variation on both. This extends Phase 1 §3.5's seed-42-vs-123 finding (different experts collapsing depending on seed) rather than just replicating it: it's not only *which* expert collapses that's seed-sensitive, it's the *total training cost and final quality* too — a real, separate cost of the current approach's variance, independent of its average performance gap.

6.6 What token-level routing actually looks like (early-training snapshot)

`scripts/expert_attribution.py` (extended this phase with Hydra CLI override support, so it can point at whatever data config a checkpoint was actually trained against — see `methodology.md` §6) was run against `hardy-lake-19`'s step-300 checkpoint. Filtering out padding (a first pass mistakenly included it — 479 of 492 positions in the sampled sequence were padding, trivially uniform) revealed real, reproducible structure, but **mostly positional, not semantic**, at this early stage: layer 0 routes only the first 1–4 tokens of every sequence to one expert (sequence-start detection) and everything else to another, regardless of topic; layer 6 shows a clean depth-progression through three experts as a sequence gets longer,

independent of content; layer 20 was fully collapsed (one expert, ~99% of all real tokens across 5 sampled sequences). One layer (7) showed a plausible content-correlated split — a block of numeric/statistical text routing differently than surrounding narrative prose — but with one sequence and no position control, this could not be distinguished from coincidence.

6.7 Which layers to select for a follow-up: a data-driven answer, this architecture's own

§7's revised recommendation points toward a *selective* layer-replacement strategy rather than all-layers. Phase 1's oracle-ceiling layer indices (10/29/30 recoverable; 17/20/31 needing cosine) don't transfer — that was a 32-layer Mistral-7B model, this is a 24-layer from-scratch one. This section derives candidates from this architecture's own data instead, combining both completed MoE runs (§6.5).

Per-layer `load_loss` , averaged across each run's full trajectory (not just the final point), both seeds:

Layer	seed=42 mean	seed=123 mean	Average	Seed-to-seed spread
0	0.223	0.206	0.214	0.017
1	0.901	1.328	1.114	0.427
2	0.769	1.701	1.235	0.932
4	1.792	1.890	1.841	0.098
17	2.344	2.159	2.252	0.185
21	3.909	3.743	3.826	0.167
20	3.328	3.601	3.464	0.273
22	4.182	3.194	3.688	0.988
23	4.264	3.307	3.786	0.957
3, 5, 9, 14, 19	—	—	mid-range	1.0–1.6 (most volatile)

(Remaining 13 layers omitted for space; the full 24-row table is reproducible from the wandb API, see §6.5's query.) Two independent signals matter here, not one: a low *average* means a layer tends to stay balanced; a low *spread* means that's reliable rather than a coin flip. Layer 0 dominates on both. Layers 4, 17, and 21 are a clear second tier — moderate-to-high averages, but consistently so (low spread) in both seeds. Layers 3/5/9/14/19 have seed-to-seed spreads of 1.0–1.6 — even where their average looks tolerable, they swing unpredictably and are poor candidates regardless of mean.

Confirmed against the final checkpoints directly, not just the training-average proxy — `expert_attribution.py` 's top-expert-share on both runs' `step_2400.pt` :

Layer	seed=42 top share	seed=123 top share
0	63.9%	58.0%
4	78.6%	91.9% (collapsed)
17	91.4% (collapsed)	97.3% (collapsed)
20	64.2%	97.9% (collapsed)

21	69.2%	97.3% (collapsed)
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Layer 0 is the only layer that stayed meaningfully balanced on *both* seeds at the final checkpoint — everything else that looked reasonable on seed=42 (4, 20, 21) collapsed hard on seed=123. This matches the training-average table's ranking exactly and is the strongest, most consistent signal in this investigation for which single layer to pick first.

What layer 0 actually learned, at the token level (padding filtered): on `graceful-universe-21` 's (seed=42) final checkpoint, layer 0 shows a clean, token-by-token split correlating with a real linguistic feature — numeric/digit tokens route to one expert, word tokens to another, consistently across all 5 sampled sequences (e.g. `2:5 3:5 @:5 .:5` vs. `the:1 city:1 population:1 was:1` in the same sequence). Layer 4 shows the same categorical split with the expert IDs swapped. The collapsed layers show either nothing (layer 17: ~everything to one expert, no structure at all) or a noisier, degraded version of the same digit/word signal (layers 20/21: mostly consistent, with real exceptions).

This does not fully reproduce on `cool-pyramid-24` 's (seed=123) final checkpoint, and the way it fails to reproduce is itself informative: layer 0 stays balanced (confirming the robustness finding above), but the *specific* boundary it draws is different — mostly one expert throughout, with a second expert taking over specifically in the numeric-dense passage plus scattered periods/conjunctions, rather than seed=42's clean per-token digit/word alternation. **The honest reading:** layer 0 reliably learns to *specialize rather than collapse* — that part is robust across seeds, confirmed three independent ways here (training-average `load_loss`, its spread, and final-checkpoint top-share). *What* it specializes on is not a fixed, guaranteed feature — the router reliably finds *some* real distinction to exploit, not always the *same* one.

Recommendation for the next run: `layers_to_replace=[0]` as the single best-evidenced candidate, with `[0, 4, 17, 21]` as a secondary option if testing more than one layer is worth the added complexity — those three are the next tier by both average and consistency, despite 4 and 21 each collapsing on one of the two seeds tested. The five volatile mid-range layers (3, 5, 9, 14, 19) should be avoided in either case; their unpredictability makes them poor building blocks for a configuration meant to be reliable.

7. Current State and What Would Be Needed Before Scaling Up

What this phase established: the infrastructure-level failure modes are found and fixed — NaN divergence, checkpoint resume, disk exhaustion, and (the big one) the initialization bug that made every prior run's absolute numbers meaningless. One full run now completes cleanly end-to-end with sane losses throughout. Now that the missing control has been run (§6.4) *and* repeated with a second seed (§6.5): **at this scale and budget, the dense baseline beats the all-layers diffusion-MoE model decisively and consistently — better `task_loss`, an order-of-magnitude-plus better `val_ppl` on both seeds, and 15-21x less wall-clock time — while also being far more seed-consistent than the MoE configuration (sub-1% run-to-run variance vs. MoE's 44-52%).**

Remaining open questions:

- The final near-zero `load_loss` coincides with the learning rate being fully decayed, on both seeds.** It is plausible routing balance was actively, robustly resolving itself in both cases — and equally plausible it simply froze wherever it happened to be as gradients vanished toward the end of the cosine schedule, which would not be evidence of a stable equilibrium at all. Two data points now both showing the same end-of-run pattern makes the "real equilibrium" reading somewhat more credible than after §6.3 alone, but doesn't settle it.
- 20M tokens is a smoke-test budget**, not a real pretraining scale, and the standard case for why sparse MoE is worth its complexity (Clark et al. 2022's routed-vs-dense scaling laws, discussed

earlier in this investigation) is specifically about quality-per-active-FLOP *at scale* — this comparison is nowhere near the regime that literature has actually demonstrated an advantage in. §6.4/§6.5's results do not show diffusion-MoE is a bad idea in general; they show this specific comparison offers no evidence for it, twice.

Recommendation, revised again: not ready for a scaled-up run, and the seed-repeat result (§6.5) removes the most likely objection to that conclusion — this was not one unlucky comparison. §8 (added after this section was first written) substantially sharpens the picture further: a selective single layer (§8.1) closes most of the all-layers gap, but the deeper cause is a moving-target problem in the diffusion router's periodic hard refit (§8.3), not sparse routing itself or this project's scale — confirmed by `SwitchMoELayer`, a structurally different (continuously-trained, no-refit) router, landing within a few percent of dense at the identical layer selection (§8.6). The practical recommendation this points to: before investing further in the diffusion-geometric router specifically, its cost (still ~70% slower than Switch even at one layer, from the CPU-bound Nyström pipeline) needs either a decisive quality advantage over Switch to justify it, or a fix to the moving-target problem itself (not just its symptoms, per §8.4) — neither of which this phase found. Switch is the stronger candidate for anything built on top of this codebase going forward unless one of those changes.

Revised once more, after §8.9: the dense baseline this whole recommendation leaned on was never compute-matched to Switch — it ran 2.7x more active FFN FLOPs/token. Compute-matched, dense and Switch tie exactly (§8.9), which doesn't change the diffusion-router verdict above (diffusion still loses to both, and by a wide margin) but does soften the case for Switch specifically: at this scale and token budget, sparse MoE routing hasn't yet shown a quality edge over dense at equal active compute, only parity at 3x the total parameter cost. Before recommending Switch-style MoE for the eventual scaled-up model (as opposed to just using it as the diffusion router's replacement in any further diffusion-specific work), the open question is whether that gap opens up in MoE's favor at a larger scale point or token budget — consistent with the standard finding in the literature that MoE's advantage tends to emerge with more data/parameters, not at this project's current 20M-token, ~sub-billion-parameter regime. Untested here.

8. Selective-Layer Results, the Refit-Discontinuity Investigation, and a Structural Alternative

Following §7's revised recommendation, this section covers what actually running §6.7's data-driven layer candidates showed, a real bug found while investigating why the routed layer wasn't recovering as expected, a deeper structural finding about *why* the recovery is incomplete even with that bug fixed, and the resulting decision to test a structurally different router as the next comparison.

8.1 Selective-layer results

Both of §6.7's candidates were run at the same 20M-token budget, `seed=42`, with the fixed initialization:

Config	task_loss	val_pp1	Wall-clock
Dense	4.80	93.2	20.1 min
Selective [0] (polar-leaf-25)	5.56	210.1	34.8 min
Selective [0,4,17,21] (smooth-donkey-26)	8.06	2,885.6	83.3 min
All-layers MoE	7.08	974.6	294.5 min

One well-chosen layer closes most of the gap to dense on every axis (`val_ppl` ~4.6x better than all-layers, wall-clock ~8.5x faster) — real support for the selective-layer direction over all-layers replacement. Adding the three second-tier candidates made things *worse*, not better — worse than `[0]` alone and worse than all-layers on `val_ppl`. This is a correction to §6.7's own framing: layers 4/17/21 were recommended based on *low seed-to-seed variance* in their `load_loss`, but their *average* `load_loss` was still moderate-to-poor (1.8–3.8, nowhere near layer 0's 0.21) — reliably mediocre is not the same as good, and diluting the one genuinely well-balanced layer with three consistently-so-so ones hurt more than it helped. Layer 0 alone remains the best MoE configuration found in this investigation.

8.2 Bug: `centroid_refresh_steps` was measured in micro-batches, not steps

Found by a sharp observation during live monitoring: a repeating loss-disruption pattern visible at 250-step intervals that didn't match the configured `centroid_refresh_steps=500` at all. Root cause:

`DiffusionMoELayer` 's internal `_step` counter increments once per `forward()` call, but `Trainer.train_step()` calls `forward()` once per *micro-batch* — `grad_accum_steps` times per logged/optimizer step, not once. Every other `*_steps` config value (`checkpoint_steps`, `eval_steps`, `log_steps`) is measured in optimizer-step units; `centroid_refresh_steps` silently wasn't. With this project's `grad_accum_steps=2` default, **every run in this investigation had been refitting landmarks twice as often as configured — 8 times over a 2,441-step run, not 4.**

Confirmed directly: the same crash-then-recover `load_loss` / `task_loss` signature already found at steps 500/1000/1500/2000 (§2.1) also appears identically at 250/750/1250/1750 in `graceful-universe-21` — exactly the halved-cadence prediction.

Fixed by threading `grad_accum_steps` into `DiffusionMoELayer` (and the full `_build_moe_layer` / `DiffusionMoETransformer` / `build_model_from_config` chain), so the refresh check is `_step % (centroid_refresh_steps * grad_accum_steps) == 0` — robust to future changes to `grad_accum_steps`, rather than requiring it to be manually doubled by hand.

8.3 The deeper finding: the model doesn't recover before the next refit hits

Investigating the disruption further (prompted by the question of whether a single refit's damage is transient or lasting) surfaced something worse than §2's original characterization. Tracing `task_loss` across a *full* refit-to-refit window in `graceful-universe-21` (step 250 to step 750):

```
step 230 (pre-refit): task_loss=6.92
step 270 (post-refit): task_loss=9.62  <- spike
step 490 (mid-window): task_loss=7.84  <- best it gets, still worse than pre-refit
step 750 (next refit): task_loss=8.45  <- never recovered, then hit again
```

`task_loss` never returns to its pre-refit level anywhere in that 500-step window, and the *next* refit lands before it can close the remaining gap. §2.1's original claim (" `load_loss` recovers within ~10-15 steps") was real but incomplete: that described `load_loss` 's aggregate *magnitude* returning to a normal noisy range, not the model's actual specialization quality recovering — the two were conflated. The honest framing: the diffusion map assumes something close to a stable manifold to fit a geometric embedding of, but the thing being embedded (post-attention activations) is itself continuously reshaped by the same gradient descent process the routing depends on (`task_loss` 's gradient flows into the attention weights that produce those activations; only the routing losses are detached — see `methodology.md` §5.2). A periodic *hard* refit repeatedly asks the router to re-orient to wherever that target has drifted to, with experts caught mid-specialization each time. This has a direct parallel in the deep-clustering literature, where alternating between fitting a clustering and updating the representation it's fit to is a known source of instability.

8.4 The sign-alignment fix helped only at the margin

A candidate mitigation: diffusion-map eigenvectors are only defined up to sign, and a refit replaces the basis outright (fresh landmarks, fresh eigensolve) with no guaranteed relationship to the old basis's orientation — `ExpertCentroids` doesn't get remapped when this happens, so an arbitrary sign flip adds a second, gratuitous discontinuity on top of the real drift from §8.3. Fixed in `NystromDiffusionMap.fit()` : snapshot the old basis's embedding of the current batch before overwriting anything, then flip each new component's sign to agree with it (confirmed against real data: a natural refit reproducibly flips a component's sign without the fix, dot product -0.0043 against the pre-refit embedding, and does not with it).

Tested head-to-head on the best selective config (`layers_to_replace=[0]` , `seed=42`): `polar-leaf-25` (before) vs. `denim-darkness-27` (after).

	Before	After
<code>task_loss</code>	5.56	5.57
<code>val_ppl</code>	210.1	211.0
Wall-clock	34.8 min	34.8 min
Peak <code>task_loss</code> in the step-250 disruption window	7.60	7.15

Essentially no change to the final outcome, though a small, real softening of the immediate disruption (peak `task_loss` 7.60 → 7.15). This is exactly what §8.3's framing predicts: sign-alignment removes a secondary source of discontinuity, but the dominant cause — the representation itself moving under the router — is untouched by it. Kept (it's a correct, cheap, structural fix, and it does measurably soften the immediate jolt even if it doesn't close the gap), but not a solution to the underlying problem on its own.

8.5 The shared expert, made a uniform toggle

Originally added only to `DiffusionMoELayer` , unconditionally on (§3). Made a `use_shared_expert` config toggle (`routing.use_shared_expert` , default `true`) threaded uniformly across all four router variants (`diffusion` / `switch` / `cosine` / `random`), so it can be compared on/off consistently rather than assumed to help everywhere, and so the comparison in §8.6 holds it constant rather than present on only one side.

8.6 A structurally different router: Switch Transformer's learned linear gate

§8.3's finding reframes the problem: the issue isn't primarily the diffusion router's refit *mechanics* (§8.2/ §8.4 both fixed real bugs there with only marginal effect) — it's that periodic hard refitting of a geometric embedding is fighting a representation that's continuously moving under gradient descent.

`SwitchMoELayer` — already implemented in this codebase as a baseline (`router: "switch"`), a single learned linear projection to expert logits, trained continuously by ordinary backprop, no diffusion map, no periodic refit, no discontinuity by construction — sidesteps this entire class of problem rather than mitigating it. Tested at the same config as the best diffusion result (`layers_to_replace=[0]` , `use_shared_expert=true` , `seed=42`):

Config	<code>task_loss</code>	<code>val_ppl</code>	Wall-clock
Dense	4.80	93.2	20.1 min

Switch [0] (efficient-hill-28)	4.93	105.3	20.2 min
Diffusion [0] (polar-leaf-25)	5.56	210.1	34.8 min
Diffusion all-layers	7.08	974.6	294.5 min

Decisive, and it answers the question this section opened with. Switch is nearly indistinguishable from dense — `task_loss` within 0.13 nats, `val_ppl` within 12, wall-clock within 6 seconds — while diffusion routing at the identical layer selection is ~2x worse on `val_ppl` and 70% slower even at its best configuration found in this investigation. This is a clean, well-controlled comparison: identical layer, identical shared-expert setting, identical seed, only the router mechanism differs. The diffusion router's problems are not primarily "sparse routing is hard at this scale" — a structurally different router (continuous, gradient-trained, no periodic refit) closes almost the entire gap to dense on its own. The periodic-hard-refit design (§8.3's moving-target problem) is the dominant cause of the diffusion router's underperformance relative to both dense and to a standard sparse-MoE alternative, not sparse routing in general and not this project's small scale.

8.7 Observability gap: wandb never logged the actual run config

Found while verifying the Switch comparison (§8.6) was actually apples-to-apples with the diffusion runs on `top_k / n_experts`: `Trainer.__init__` called `wandb.init(project=...)` with no `config=` argument at all, for every run in this entire investigation. None of a run's actual hyperparameters — `routing.top_k`, `model.router`, `use_shared_expert`, any of it — were ever recorded in wandb. Confirming what a given run actually used required SSHing into the instance and reading its log file by hand (and even that was often unavailable in the moment: `nohup` 's stdout redirection is block-buffered, so the printed config frequently hadn't flushed to disk yet for a still-running process — the same buffering behavior noted throughout this investigation's live-monitoring sections).

Fixed: `wandb.init(..., config=wandb_config)`, where `wandb_config` is `OmegaConf.to_container(config, resolve=True)` for a real Hydra `DictConfig` (flattened into a plain, JSON-serializable dict — passing the `DictConfig` object through as-is would not have logged correctly), falling back to the config object as-is for the plain-dict configs the test suite uses. Every run from this point forward has its actual config in wandb's own config panel — no more reconstructing it from launch commands or SSH sessions after the fact.

8.8 Switch generalizes: all-layers scaling and seed-robustness, both confirmed

Two follow-up runs, both `use_shared_expert=true`, `top_k=2`:

Config	<code>task_loss</code>	<code>val_ppl</code>
Dense	4.80	93.2
Switch, all layers (faithful-jazz-29, seed=42)	4.87	100.4
Switch [0] (efficient-hill-28, seed=42)	4.93	105.3
Switch [0] (winter-moon-31, seed=123)	4.90	104.9
Diffusion [0] (polar-leaf-25)	5.56	210.1
Diffusion all-layers (graceful-universe-21)	7.08	974.6

Two results, both favorable to Switch and both a sharp contrast with diffusion's behavior at the same two axes:

- **Scaling to all layers helps Switch, not hurts it.** All-layers Switch (`val_ppl` 100.4) is closer to dense (93.2) than layer-0-only Switch is (105.3) — the opposite of diffusion, where going from one layer (210.1) to all layers (974.6) made things roughly 4.6x worse. Switch's continuously-trained linear gate doesn't accumulate the periodic-refit disruption (§8.3) that compounds across layers for the diffusion router; more routed layers is just more useful sparse capacity.
- **Seed-robustness matches dense, not diffusion.** `efficient-hill-28` (seed=42) and `winter-moon-31` (seed=123) land within 0.6% of each other on `task_loss` and 0.4% on `val_ppl` — comparable to dense's sub-1% seed variance (§6.5), and nowhere near diffusion's 44-52% seed-to-seed swings at the same layer selection.

Together with §8.6, this closes out the comparison cleanly across every axis tested — single-layer quality, all-layers scaling, and seed stability — and all three favor Switch over the diffusion router by a wide, consistent margin.

A real gap worth flagging, then checked empirically. Neither load-balance loss in this codebase (`coefficient_of_variation_loss` , `switch_load_balance_loss`) penalizes anything beyond marginal per-expert dispatch frequency — a router that consistently pairs experts into redundant duplicate groups would score as perfectly "balanced" under either loss, since both are satisfied by symmetric duplication just as well as by genuine diversity. No orthogonality/diversity term exists anywhere in this codebase, across any of the four router variants. Checked directly against the one surviving Switch checkpoint (`winter-moon-31` , layer 0, `n_experts=8` , `top_k=2` — the `checkpoints/` directory is overwritten per-run, so `faithful-jazz-29` 's all-layers checkpoint is no longer available) using `scripts/expert_attribution.py` plus a direct weight comparison:

- **Token specialization:** each expert's top-15 routed tokens (pooled over 20 held-out batches) are visibly distinct — e.g. expert 1 is dominated by . and capitalized proper-noun starts (`P / W / D / G`), expert 4 by function words (`the / of / in / by`), expert 3 by numerals and was . Mean pairwise Jaccard overlap across all 28 expert pairs' top-token sets: 0.08 (range 0.00-0.21) — far from the ~1.0 two truly redundant experts would show.
- **Weight-space similarity:** pairwise cosine similarity between experts' `down_proj` weight matrices is statistically indistinguishable from a random-initialization baseline (trained: mean -0.0005, max 0.007; random-init baseline of the same shape: mean 0.0002, max 0.003) — no anomalous alignment between any pair.
- **Load:** fairly even in training (`load_loss` ~0 for all three §8.6/§8.8 Switch runs), though the held-out attribution sample shows real spread (1,045-3,974 tokens across the 8 experts, 3.8x) — not collapsed, but not perfectly uniform either.

For this run, the duplication concern doesn't appear to be materializing — the 8 experts look genuinely differentiated by both routing outcome and learned weights, despite no explicit mechanism preventing it. This isn't proof it can never happen (only one checkpoint checked, and weight-cosine in a ~500K-dim space is a weak test for redundancy expressed via internal hidden-unit permutation rather than raw parameter alignment — a functional-output-correlation check on shared activations would be the more rigorous version if this needs revisiting), but it's no longer a purely theoretical worry left unchecked.

(Note: the two follow-up runs launched back-to-back initially collided on the single-GPU instance — the second crashed with a CUDA OOM while the first was still resident — and had to be resequenced. Unrelated to the architecture; noted here only because it's a recurring operational hazard on a single-GPU box.)

8.9 The dense-vs-Switch comparison was never compute-matched — corrected

Every dense-vs-MoE comparison so far (§6.4, §6.5, §8.6, §8.8) used `model.ffn_dim=4096` for dense (the config default) against Switch's actual active per-token FFN width, without checking whether those were comparable in the first place. They weren't: ExpertFFN 's hidden width is `ffn_dim // n_experts × overlap_factor = 4096 // 8 × 1.0 = 512` , and each token activates `top_k` routed experts plus the always-on shared expert (§3), so Switch's true active FFN compute per token is `(top_k + 1) × 512 = 1,536` — dense at `ffn_dim=4096` was running **2.7x more active FLOPs/token** than Switch the entire time. The 93.2-vs-100.4 `val_ppl` gap in §8.8 partly (perhaps entirely) reflected that compute disparity, not an architectural advantage for dense.

Corrected: a dense run with `model.ffn_dim=1536` , exactly matching Switch's active FFN width, same data/budget/seed as every other run in §8:

Config	Active FFN FLOPs/token	Total FFN params/layer	task_loss	val_ppl
Dense, <code>ffn_dim=4096</code> (uncorrected baseline)	12.6M	12.6M	4.80	93.2
Dense, <code>ffn_dim=1536</code> (wild-eon-32, compute-matched)	4.7M	4.7M	4.877	100.33
Switch, all layers (faithful-jazz-29)	4.7M	14.2M (3x dense)	4.871	100.4

At matched active compute, dense and Switch are statistically indistinguishable (`val_ppl` 100.33 vs 100.4, `task_loss` 4.877 vs 4.871 — well within this project's measured seed-to-seed noise floor, §6.5/§8.8). Switch carries 3x the total FFN parameters of compute-matched dense for that tie, not a win — the extra routed capacity isn't translating into a quality advantage at this scale/token budget. This is a materially different conclusion from §8.6/§8.8's framing: Switch isn't "nearly as good as dense," it's *exactly* as good as a dense model given the same compute, and the sparse-MoE structure isn't yet earning its added complexity and 3x parameter cost. Whether that changes at a larger token budget or model scale (where MoE's literature advantage typically shows up) is untested — this is one data point, not a scaling curve.

8.10 Literature context: is this tie a novel finding, or the expected one?

`SwitchMoELayer` is a faithful (if simplified) implementation of Fedus, Zoph & Shazeer 2021 ("Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity") — the same learned-linear-gate-plus-load-balance-loss design underlies most production sparse-MoE LLMs since (GShard, GLaM, Mixtral, DeepSeek-MoE). §8.9's tie is consistent with, not contradicted by, the literature on this architecture:

- The field's routed-vs-dense quality advantage is itself scale-dependent (Clark et al. 2022, "Unified Scaling Laws for Routed Language Models" — already the source cited in §7 for why a single small-scale comparison shouldn't be over-trusted either way). MoE's advantage over compute-matched dense is a claim about the large-scale asymptote, not guaranteed to hold at every budget.
- MoE models are typically *more* data-hungry per active parameter than dense at a fixed compute budget (Krajewski et al. 2024, "Scaling Laws for Fine-Grained Mixture of Experts"), meaning an under-trained small-budget comparison like this project's 20M tokens is exactly the regime where MoE would be expected to underperform its asymptotic potential, not a neutral test of it.
- This implementation is also missing two components the literature found necessary for MoE to reach its potential at scale: an **expert capacity factor** (the original Switch Transformer caps tokens routed to each expert per batch and drops overflow, which both bounds compute and forces real

load balancing under a hard constraint) and a **router z-loss** (added in the ST-MoE follow-up, Zoph et al. 2022, specifically to stabilize router logits; found to measurably improve quality). Neither exists anywhere in this codebase (confirmed by `grep` across `src/`) — plausible, untested candidates for narrowing the tie into a real Switch advantage even at this project's current scale, before reaching for a bigger scale point.

Net read: §8.9's tie is the expected outcome given the literature, not a surprising negative result about this specific implementation — and it points at concrete, cheap next experiments (capacity factor, z-loss) before concluding anything about whether MoE "works" here at all.

9. Code Changes Summary

File	Change
<code>geometry/nystrom.py</code>	Guarded the <code>d_alpha_new</code> division against exact-zero underflow; added live <code>eps_landmark-spectrum</code> refresh in <code>transform()</code> (new <code>_refresh_eps</code> param, default on); extracted <code>_fit_landmark_spectrum</code> shared between <code>fit()</code> and the refresh path; <code>fit()</code> now sign-aligns the freshly-fit eigenbasis to agree with the previous one's embedding of the same batch (§8.4)
<code>geometry/eigensolver.py</code>	<code>diffusion_eigenvectors</code> gained a <code>random_state</code> param, seeding ARPACK's starting vector — needed once the eigensolve could rerun mid-training, not just once per landmark set
<code>models/moe_layer.py</code>	Added <code>shared_expert</code> (outside the router; §3); <code>_compute_diffusion_coords</code> now also refits when <code>ndm.landmarks_</code> is <code>None</code> (checkpoint-resume fix), no only on the step-count schedule; refresh cadence now accounts for <code>grad_accum_steps</code> (§8.2); <code>use_shared_expert/grad_accum_steps</code> constructor params
<code>models/switch_moe_layer.py</code> , <code>models/cosine_moe_layer.py</code> , <code>models/random_moe_layer.py</code>	Added the same <code>use_shared_expert</code> toggle (§8.5) — the shared expert is no longer <code>DiffusionMoELayer</code> -only
<code>models/moe_model.py</code>	Fixed <code>token_embedding</code> init (<code>std=0.02</code> , was <code>nn.Embedding</code> default <code>std=1.0</code>); added depth-aware <code>1/sqrt(2×n_layers)</code> scaling to every <code>out_proj/down_proj</code> weight after construction; threads <code>grad_accum_steps/use_shared_expert</code> through <code>_build_moe_layer</code>
<code>models/model_factory.py</code>	Reads <code>training.grad_accum_steps</code> and <code>routing.use_shared_expert</code> from config
<code>training/trainer.py</code>	Added non-finite-loss guard in <code>train_step</code> (raises <code>FloatingPointError</code> before <code>backward()/optimizer.step()</code> instead of silently training on <code>NaN/Inf</code>); added <code>_prune_old_checkpoints</code> (<code>keep_last_n_checkpoints</code> , default 2); <code>wandb.init</code> now actually passes <code>config=</code> (§8.7) — was never logging any run's real hyperparameters at all
<code>data/data_loader.py</code>	Fixed <code>local_data_files</code> reaching the real Hydra-config-driven training path — a <code>ConfigAttributeError</code> several frames deep (<code>omegaconf.ListConfig</code> vs. plain <code>list</code>), not an obvious error at the call site

scripts/expert_attribution.py	Added Hydra-style CLI override support (parse_known_args), so it can point at the real data config a checkpoint was trained against instead of always falling back to base_config.yaml's defaults
scripts/train.py	cfg.seed now actually applied (torch.manual_seed/np.random.seed/random.seed/torch.cuda.manual_seed_all before model construction — previously a no-op
configs/base_config.yaml	routing.mu_load: 0.01 → 2.0 (measured, §6.2); training.keep_last_n_checkpoints: 2 (new); routing.use_shared_expert: true (new, §8.5)

All changes are covered by tests reproducing the specific failure before confirming the fix, not just testing the fix in isolation — e.g. a mixed-batch outlier reproducing the exact NaN-underflow condition; a cross-object checkpoint resume at a step deliberately off the refit schedule, confirmed to crash on the pre-fix code (verified by temporarily reverting the fix) and pass after; a monkeypatched zero-contribution routed path confirming the shared expert's contribution is structurally unconditional; a real disk-bounded checkpoint-pruning scenario; the random-label cross-entropy check confirming init-time loss lands near the true random baseline (confirmed to fail — 61.5 vs. an expected <13.8 — on the pre-fix code); a same-seed-reproduces/different-seed-diverges check on model initialization; a centroid_refresh_steps / grad_accum_steps interaction check confirming the correct (not halved) refit cadence; and a sign-alignment check confirming a real refit reproducibly flips a component's sign without the fix and does not with it. Full suite: 385 tests passing at last check.

10. Phase Closed: Final Status

This phase is closed as of this report. Summary of where each track landed, and what would justify reopening it:

Diffusion-geometric router — on hold, needs a structural fix, not further tuning. Decisively behind both dense and Switch on every axis tested this phase: quality (§6.4, §8.1, §8.6), cost (§8.6 — ~70% slower than Switch even at one layer), and seed-robustness (§6.5, 44-52% variance vs. Switch/dense's sub-1%). Root-caused, not just observed: the periodic hard refit is fighting a representation continuously reshaped by the same gradient descent the routing depends on (§8.3), and a targeted fix for the coordinate-discontinuity symptom (§8.4, sign-alignment) helped only at the margin — confirming the problem is the moving-target dynamic itself, not refit mechanics. **What would reopen this track:** a real structural redesign (e.g. a soft/continuous update to the geometric embedding instead of a periodic hard refit, or detaching fewer signals from the graph so the embedding target moves less) — not another round of hyperparameter or refit-cadence tuning, both of which this phase already tried.

Switch-style sparse MoE — ties compute-matched dense here; advantage over dense not yet demonstrated at this scale. Nearly matches dense at a naive (uncorrected) comparison (§8.6, §8.8) and ties it exactly once compute is actually matched (§8.9) — at 3x the parameter cost for that tie. Consistent with the literature's expectation that MoE's edge is scale-dependent and this project's 20M-token budget is smaller than where that edge typically shows up (§8.10), and this implementation is missing two components (expert capacity factor, router z-loss) the literature found relevant at scale. **What would reopen this track:** either a cheap in-place experiment (capacity factor and/or z-loss at the current scale, per §8.10) or a genuine second scale point (larger token budget/model size) to see whether the tie breaks in Switch's favor as literature would predict — this phase did not run either.

What this phase leaves behind, concretely: a working all-layers training pipeline (seven real bugs fixed, §9), a data-driven layer-selection methodology (§8.1/§6.7) and a compute-matching methodology (§8.9/methodology.md §6.6) reusable for any future comparison on this codebase, and a duplication/separation-checking methodology (§8.8/methodology.md §6.7) for auditing any router's learned experts. Nothing here is a dead end — both open tracks have a specific, named next step, not just "try harder."

Appendix: Open Questions Carried Forward

- Does the depth-progression / sequence-start routing pattern (§6.6) persist, sharpen into content-sensitivity, or dissolve entirely at a later training step and a larger token budget? **Partially answered for layer 0 by §6.7:** by the final checkpoint it had moved past pure position into a real content-correlated split (digit vs. word tokens on one seed), though the specific feature wasn't identical on the other seed. Layer 6 specifically (the depth-progression layer) wasn't re-checked at the final checkpoint — still open. (Diffusion-specific; low priority given §10's disposition on that track.)
- Should `nu_sep` (currently unchanged at 0.05) be measured the same way `mu_load` was (§6.2)? `sep_loss` showed real instability of its own (swings from -3.5 to -10.6 within ~120 steps in the collapsed run) that was not directly investigated — it may be a downstream symptom of the same collapse dynamic, or a separate miscalibration. (Diffusion-specific; low priority given §10's disposition on that track.)
- The self-tuning (per-point adaptive) bandwidth design discussed but not implemented (the "Option 2" alternative to §2.2's global live-refresh) remains a candidate if staleness-driven instability recurs even with the current fix. (Diffusion-specific; low priority given §10's disposition on that track.)
- `NystromDiffusionMap`'s own k-means/eigensolve randomness still has a hardcoded `random_state=42`, independent of `cfg.seed` (§9) — the seed-repeat check in §6.5 varied model init and data-adjacent RNG state, but not this. Worth wiring through too if the seed-sensitivity finding in §6.5 needs isolating further (is it driven by init, by the diffusion-map's own landmark selection, or both). (Diffusion-specific; low priority given §10's disposition on that track.)
- **Switch expert-capacity factor and router z-loss (§8.10)** — neither implemented; both plausible candidates for closing the compute-matched tie (§8.9) in Switch's favor without needing a larger scale point. Highest-priority open item if this track is reopened.
- **A second scale point for Switch vs. dense (§8.10, §7)** — the compute-matched tie (§8.9) is one data point at one (small) budget; the literature's MoE-advantage claim is about the large-scale asymptote specifically. Untested here.
- **Functional-output-correlation check for expert duplication (§8.8, methodology.md §6.7)** — the token-overlap and weight-cosine checks both have known blind spots (internal hidden-unit permutation invariance in particular); a shared-activation output-correlation check would be the more rigorous version if the duplication question needs revisiting with higher confidence than one checkpoint currently supports.